Human-AI Interaction:

Human Multisensory Perception

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Motivation

- Humans use multiple modalities to communicate (e.g., speech, facial expressions, gestures).
- We sense the world through multiple channels — not just five.
- This sub-course consists of two parts:
    - **Multisensory perception**: how humans sense and interpret the world through our senses combine these channels.
    - **Multimodal processing**: how to design AI systems that can process and integrate information from multiple modalities.

Overview: Sensation, Perception & the Senses

Topic
Sensation vs. Perception
Waves and Wavelengths
Vision
Hearing
The Other Senses
Gestalt Principles of Perception

Sensation vs. Perception

Sensation — occurs when sensory receptors detect stimuli and convert them into neural signals sent to the brain.

Transduction — conversion from sensory stimulus energy to action potential.

Stimuli (sing. *stimulus*) — any detectable change in the internal or external environment.

Perception — the way sensory information is organized, interpreted, and consciously experienced.

Our Sensory Modalities

More than five senses exist:

Sense	Modality
Vision	Light
Hearing (audition)	Sound waves
Smell (olfaction)	Chemical
Taste (gustation)	Chemical
Touch (somatosensation)	Mechanical
Balance	Vestibular
Body position/movement	Proprioception & Kinesthesia
Pain	Nociception
Temperature	Thermoception

Absolute Threshold

Absolute threshold — minimum stimulus energy detectable 50% of the time.

Examples:

- Candle flame visible from **30 miles** away on a clear night
- Clock tick audible from **20 feet** away in quiet conditions

Subliminal messages — stimuli below threshold for conscious awareness. Research shows little effect on real-world behavior.

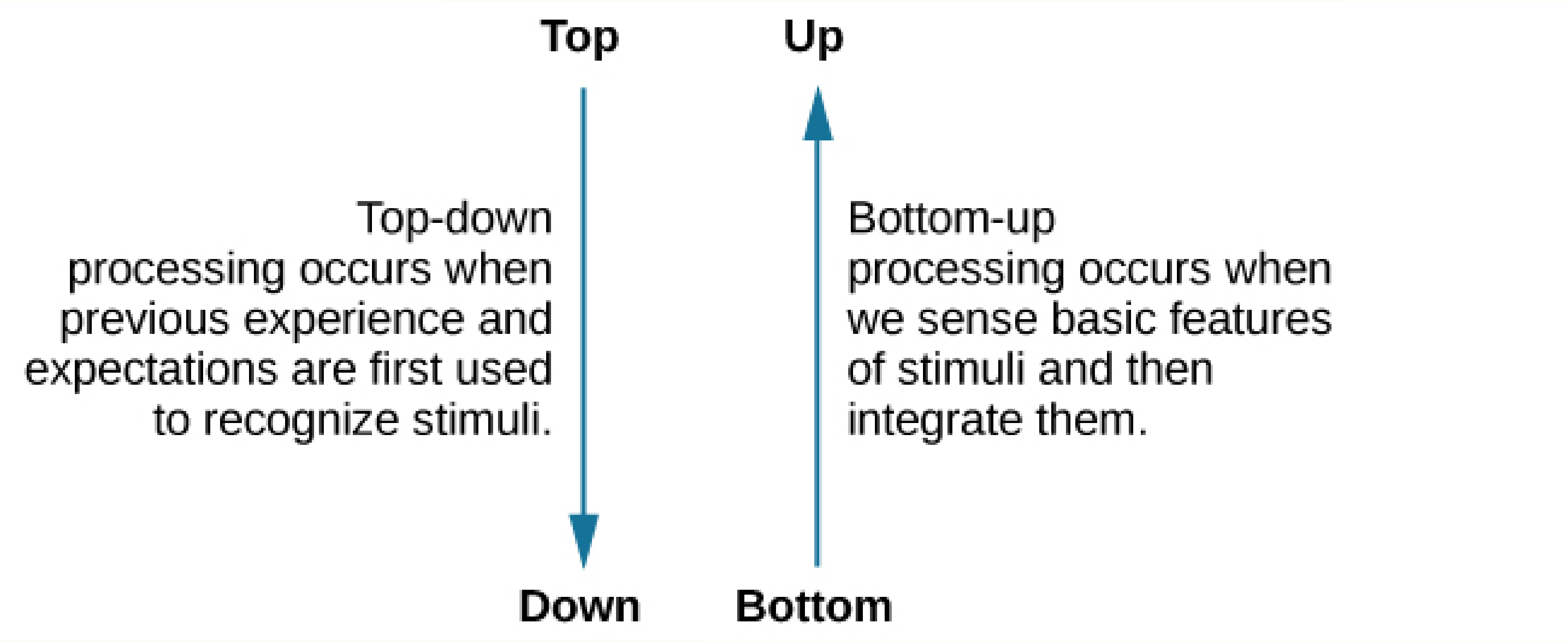
Difference Threshold & Weber's Law

Just noticeable difference (JND) — the smallest detectable difference between two stimuli.

Weber's Law: The difference threshold is a *constant fraction* of the original stimulus.

Example: A phone screen lighting up is obvious in a dark movie theater, but unnoticed in a brightly lit arena — same brightness, different context.

Bottom-Up vs. Top-Down Processing



Top-down processing uses prior knowledge; bottom-up processing is driven by sensory input.

Bottom-Up & Top-Down

Bottom-up: automatic, stimulus-driven, involuntary

- e.g., crashing sound in a restaurant captures attention regardless of what you were focusing on

Top-down: goal-directed, deliberate, voluntary

- e.g., scanning for a yellow key fob in likely locations, ignoring the ceiling fan

Sensation is **physical**; Perception is **psychological**.

Sensory Adaptation

Sensory adaptation — the decrease in sensitivity to a stimulus that remains constant over time.

Example: A flashing construction light outside your hotel window seems very annoying at first, but after watching TV for a while, you no longer notice it — even though your photoreceptors still detect it.

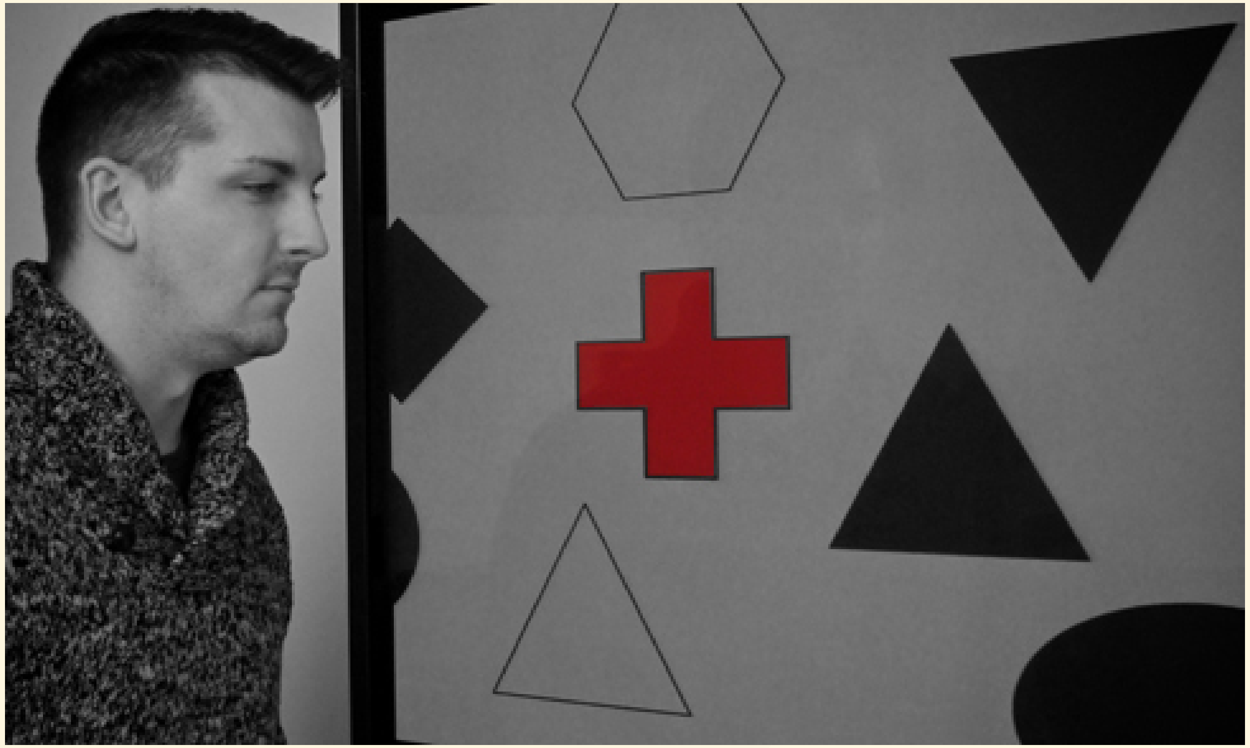
Attention & Inattentional Blindness

Inattentional blindness — failure to notice a fully visible stimulus because attention is elsewhere.

Simons & Chabris (1999): While counting basketball passes, nearly **half** of participants failed to notice a gorilla walking through the scene — visible for 9 seconds.

Signal detection theory — ability to identify a stimulus embedded in background noise; affected by motivation and expectation.

Inattentional Blindness

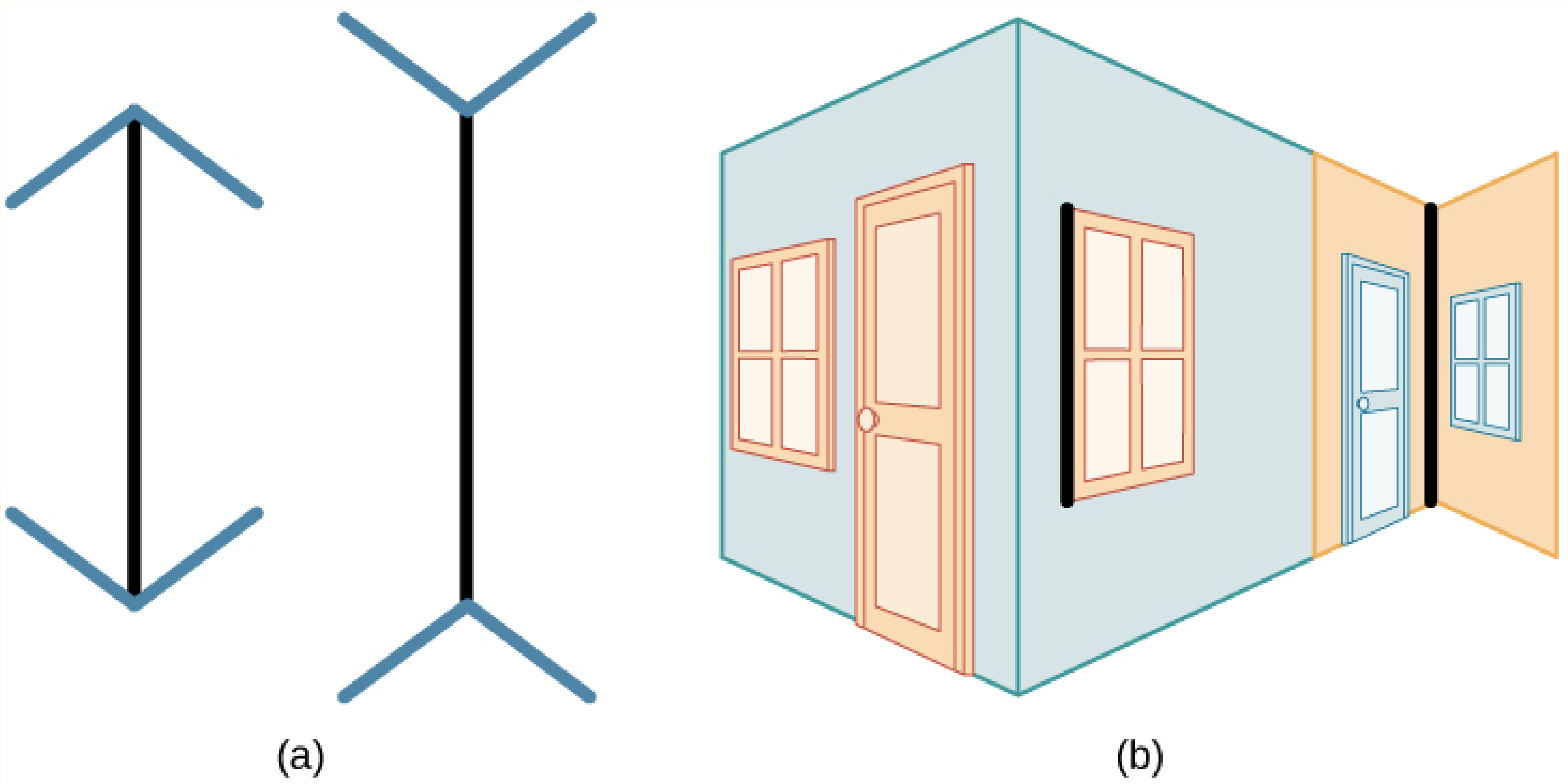


About ⅓ of participants missed a red cross on screen because attention was focused on black/white figures. (credit: Cory Zanker)

Cultural & Individual Effects on Perception

- Beliefs, values, personality, and culture all shape perception.
- Segall et al. (1963): Western cultures (carpentered world) more susceptible to the **Müller-Lyer illusion**; Zulu people (round structures) less susceptible.
- Cross-cultural differences also found in odor identification and taste preferences.

Müller-Lyer Illusion



Lines appear to be different lengths although they are identical — a classic example of how top-down expectations shape perception.

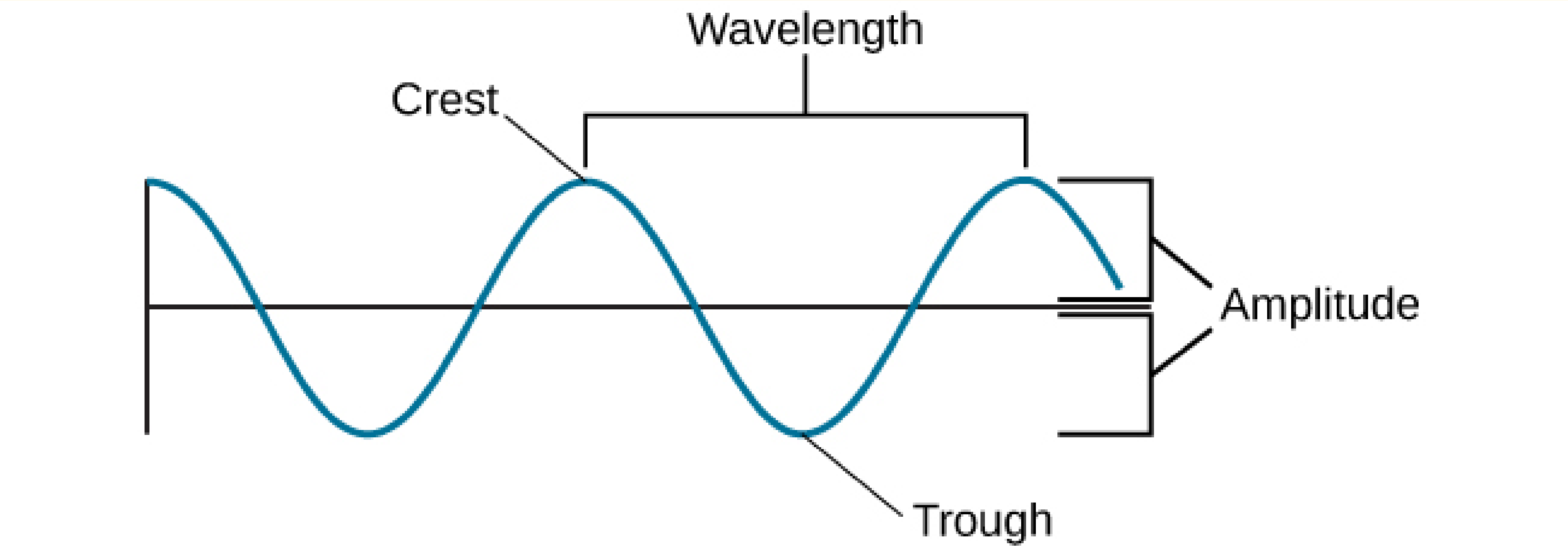
Waves and Wavelengths

The key physical characteristics of a wave:

- **Amplitude** — distance from center line to crest (peak) or trough; associated with **intensity**
- **Wavelength** — length from one peak to the next; inversely related to frequency
- **Frequency** — number of wave cycles per second, measured in **hertz (Hz)**

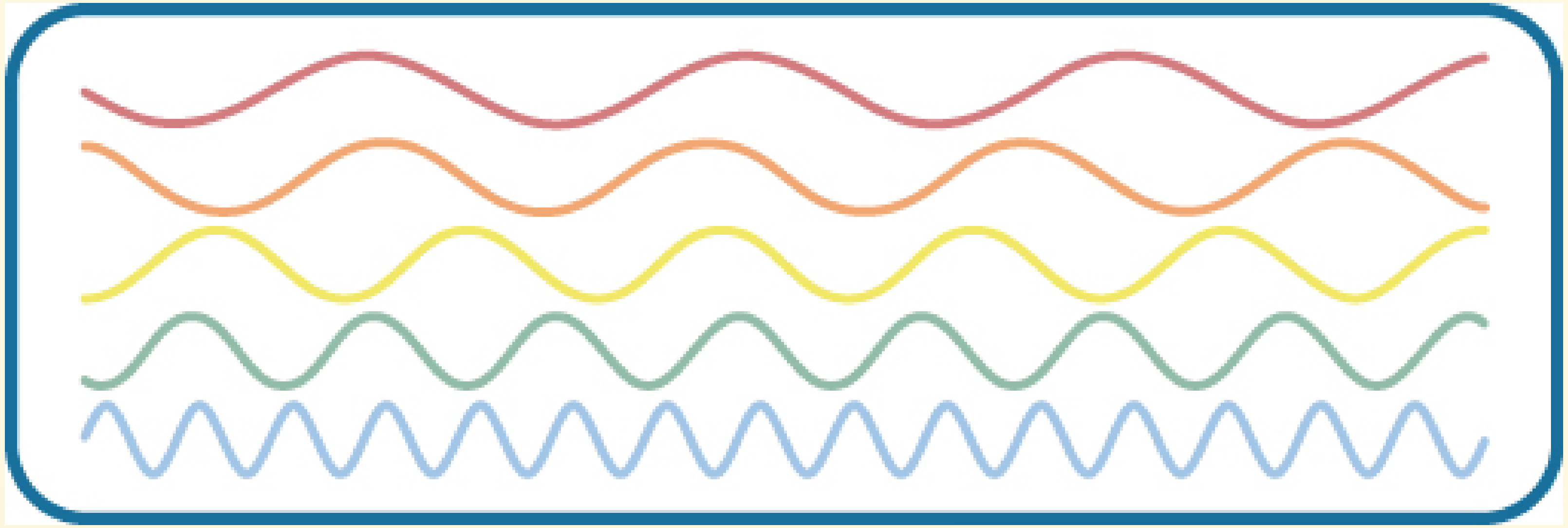
Higher frequency = shorter wavelength; lower frequency = longer wavelength.

Wave Amplitude and Wavelength



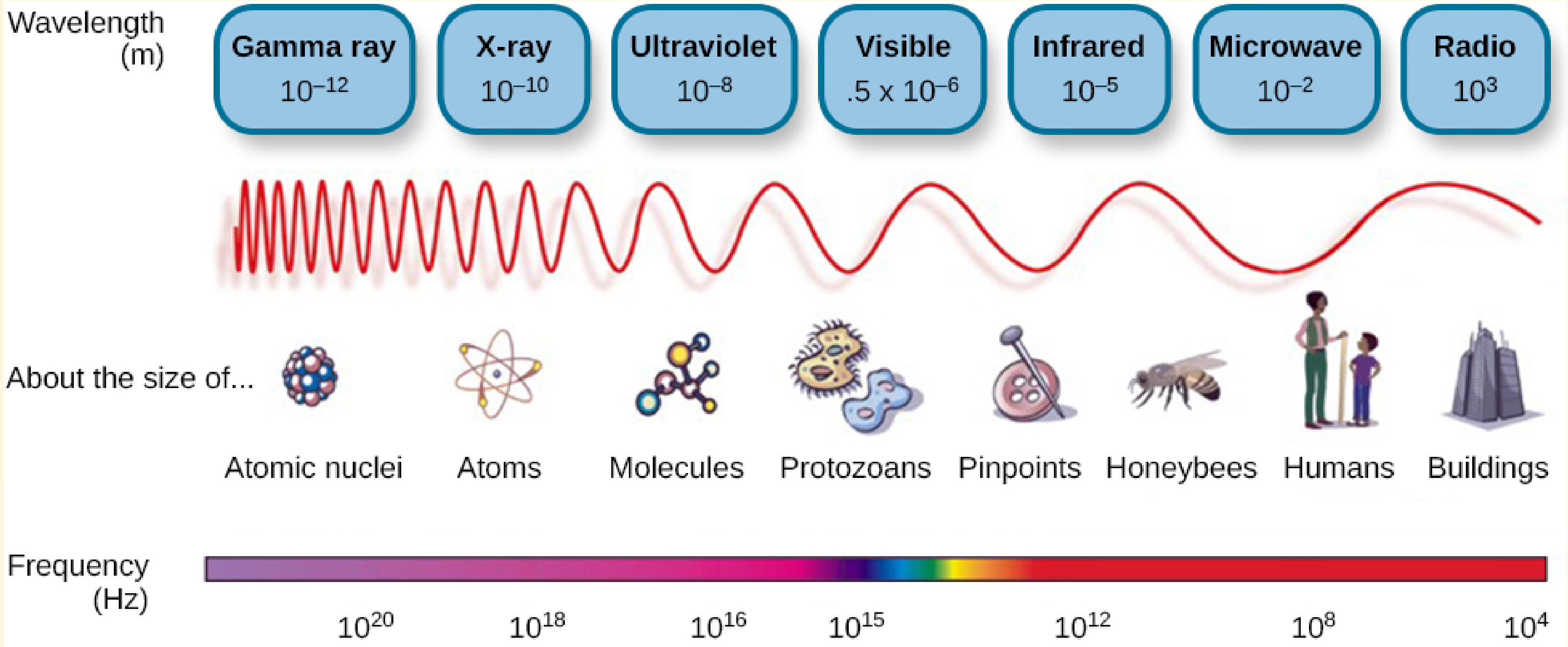
Amplitude is measured from peak to trough; wavelength is measured peak to peak.

Frequency and Wavelength



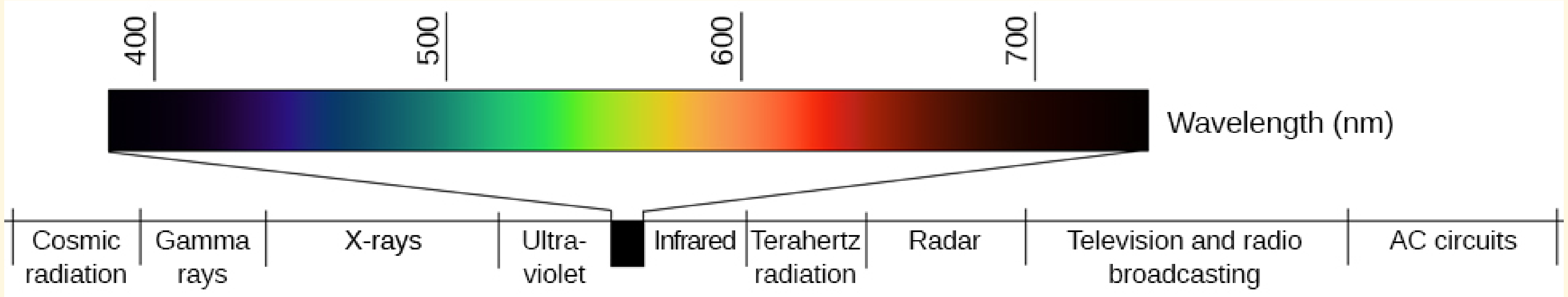
Waves of differing wavelengths and frequencies. From top to bottom: wavelengths decrease, frequencies increase.

Light Waves: The Electromagnetic Spectrum



Visible light is only a small portion of the full electromagnetic spectrum (380–740 nm). Other species detect beyond human range: honeybees (UV), some snakes (infrared).

Color and Wavelength

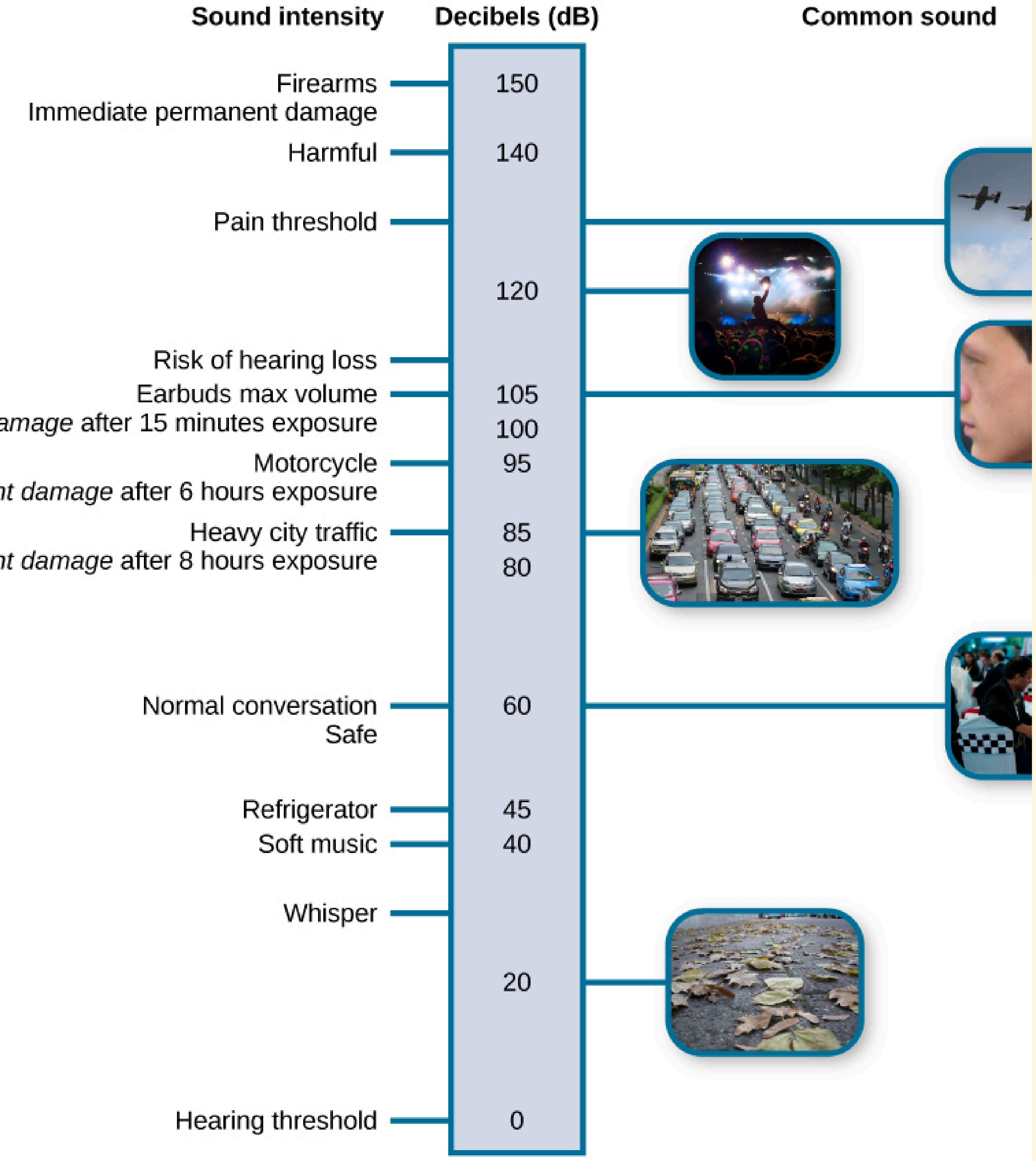


Different wavelengths correspond to different colors. Mnemonic: **ROYGBIV** (Red, Orange, Yellow, Green, Blue, Indigo, and Violet).

- Reds → longer wavelengths
- Violets/Blues → shorter wavelengths
- Amplitude → brightness/intensity

Sound Waves

- **Pitch** ↔ frequency: high frequency = high pitch; low frequency = low pitch
- Human audible range: **20–20,000 Hz**
- **Loudness** ↔ amplitude, measured in **decibels (dB)**
  - Conversation ≈ 60 dB; Rock concert ≈ 120 dB
  - Hearing damage risk from ~80 dB; pain threshold ~130 dB
- **Timbre** — a sound's purity; explains why instruments sound different at same pitch and loudness



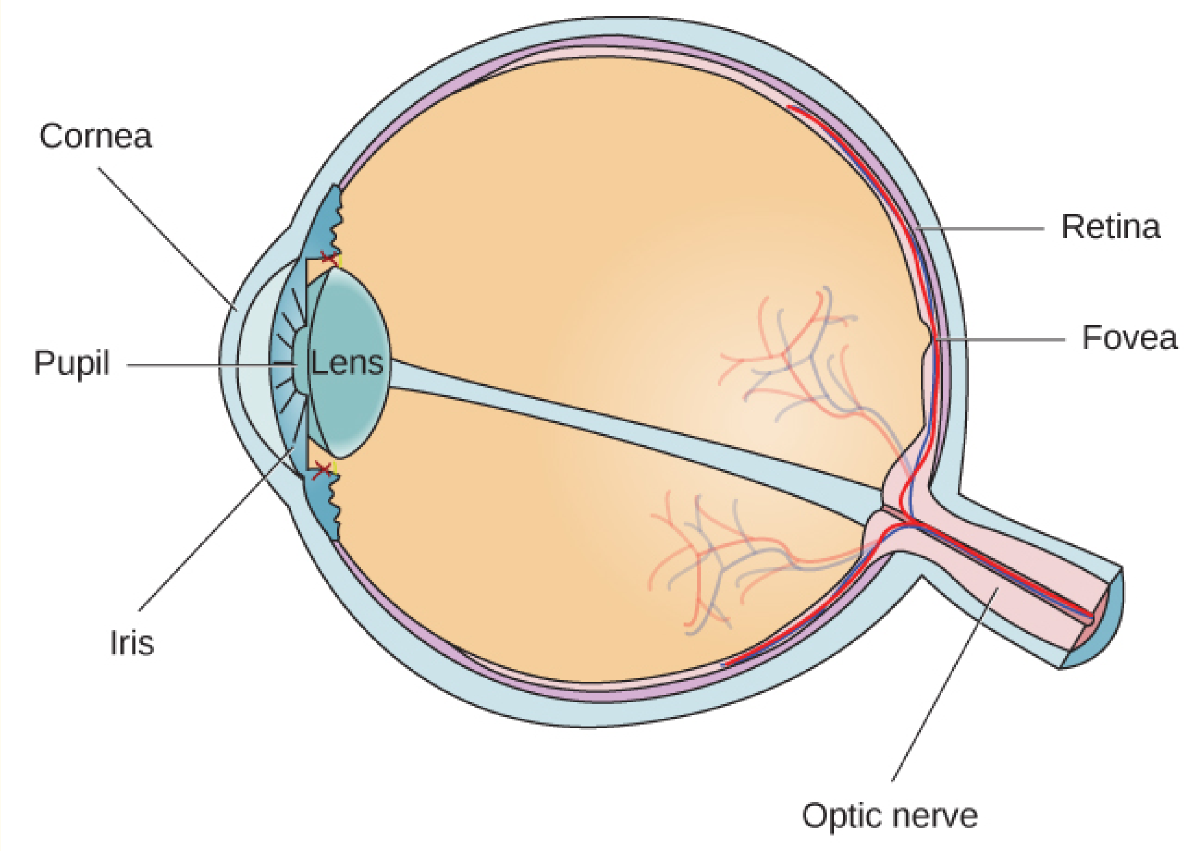
Loudness of Common Sounds

Loudness levels of common sounds: ~⅓ of all hearing loss is due to noise exposure.

Vision: The Eye

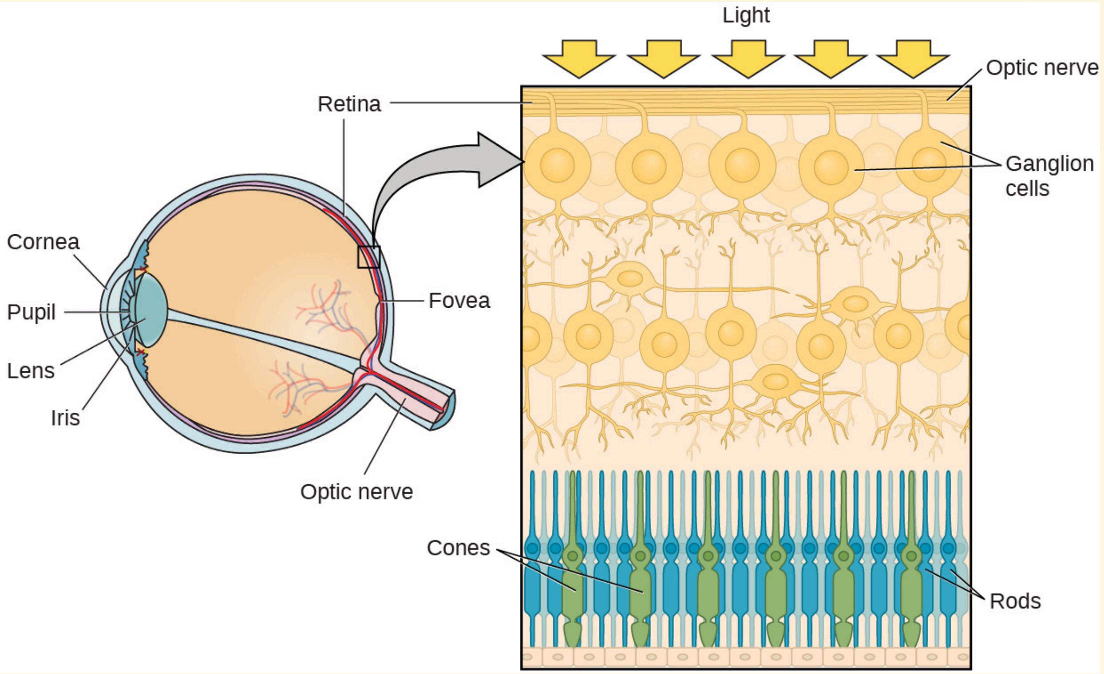
Our eyes gather sensory information for interpreting the world.

Key structures:
cornea → pupil → lens → retina → fovea → optic nerve

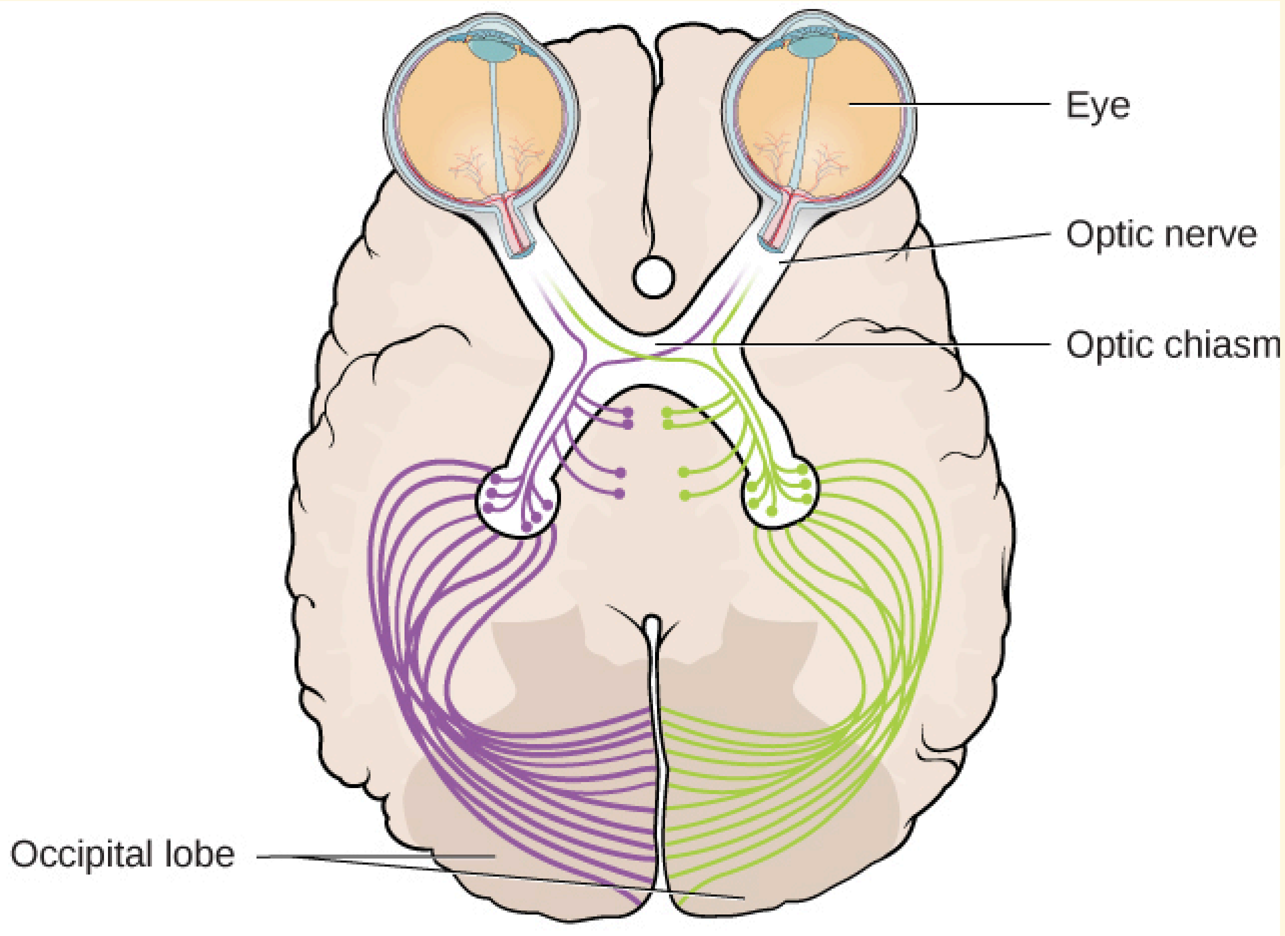


Photoreceptors: Rods and Cones

Cones (green) are concentrated in the fovea; rods (blue) cover the rest of the retina.



	Rods	Cones
Location	Periphery	Fovea
Light condition	Dim	Bright
Function	Motion, low-light	Color, detail



At the optic chiasm, right-visual-field info goes to left hemisphere and vice versa. Processed in occipital lobe.

- **"What" pathway** – object recognition
- **"Where/How" pathway** – location and spatial interaction

Color Vision

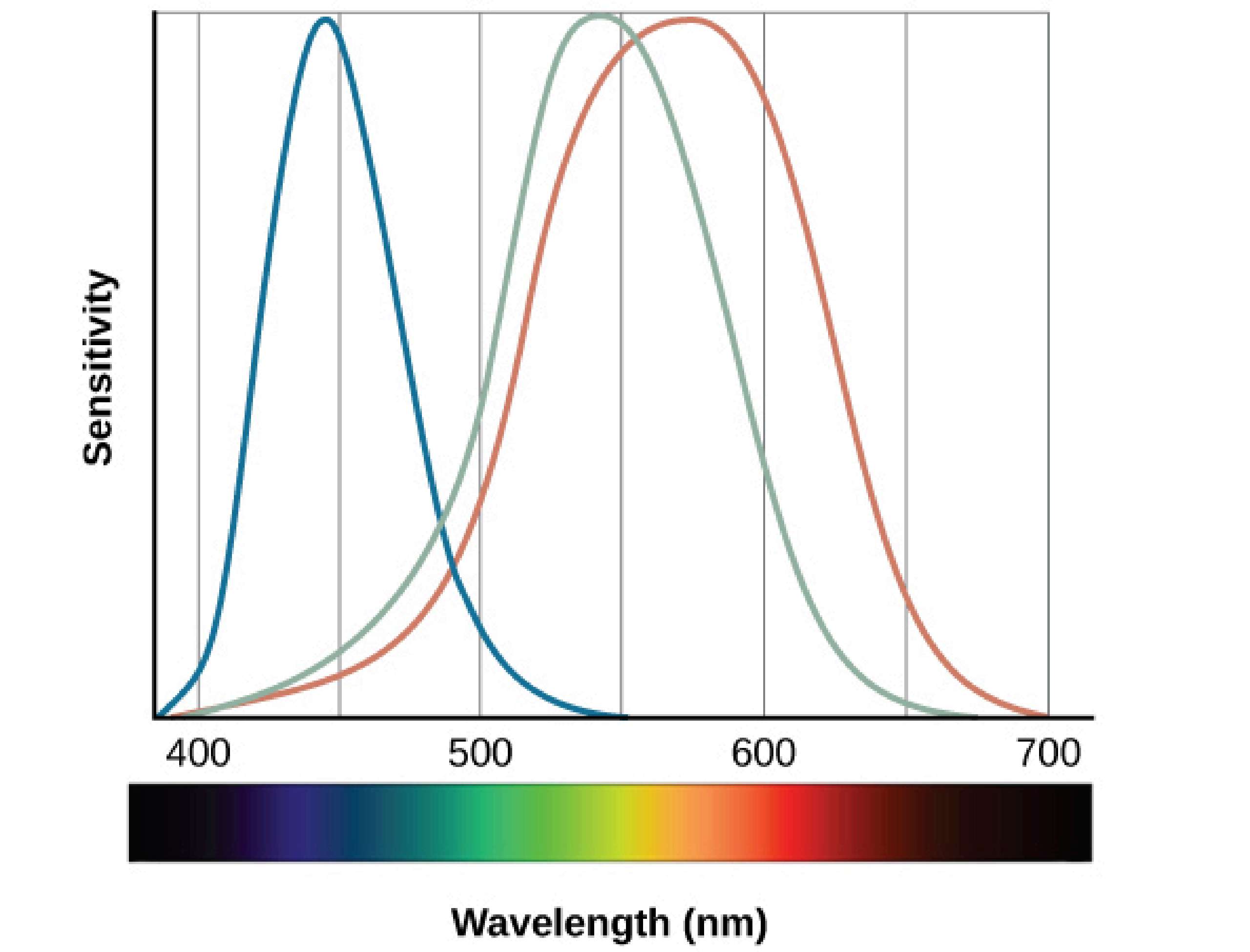
Trichromatic theory: three cone types sensitive to red, green, and blue; all colors produced by combining these.

Opponent-process theory: color coded in opponent pairs (red-green, blue-yellow, black-white). Explains **negative afterimages**.

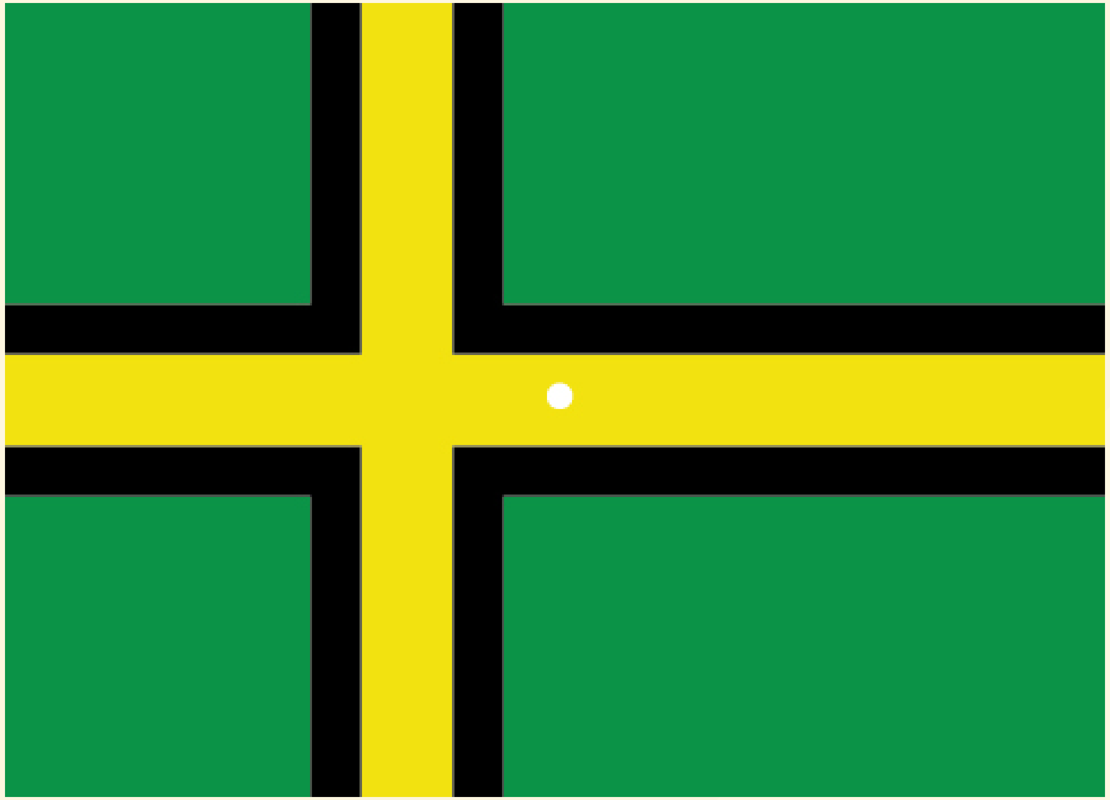
Both theories apply — trichromatic at the retina level; opponent-process further up toward the brain.

Trichromatic Theory

Sensitivity curves for the three cone types. Red-green color blindness affects ~8% of European males.



Negative Afterimage



Stare at the white dot for 30–60 seconds, then look at a white surface. The afterimage demonstrates opponent-process theory.

Depth Perception

Binocular cues (both eyes):

- **Binocular disparity** — slightly different view from each eye; basis of 3-D movies

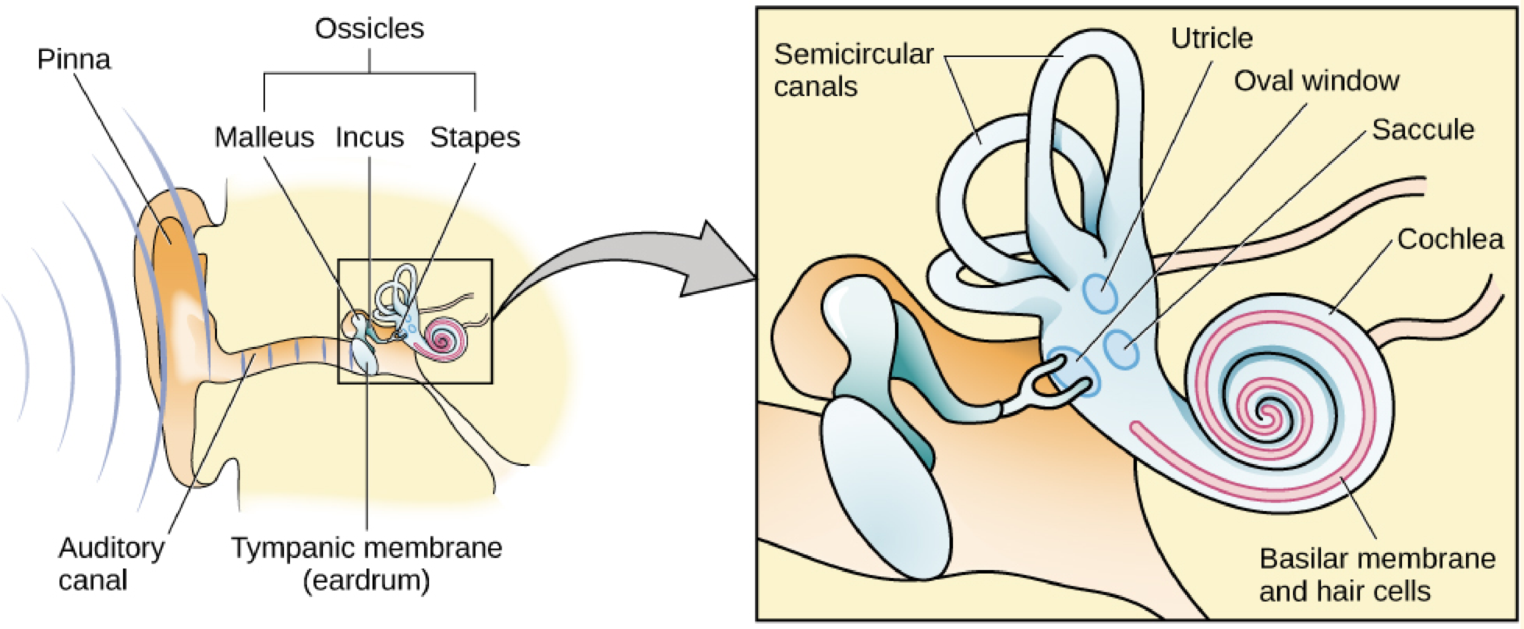
Monocular cues (one eye):

- **Linear perspective** — parallel lines converge in the distance
- **Interposition** — partial overlap of objects
- Relative size, closeness to horizon

Linear Perspective (Monocular Depth Cue)

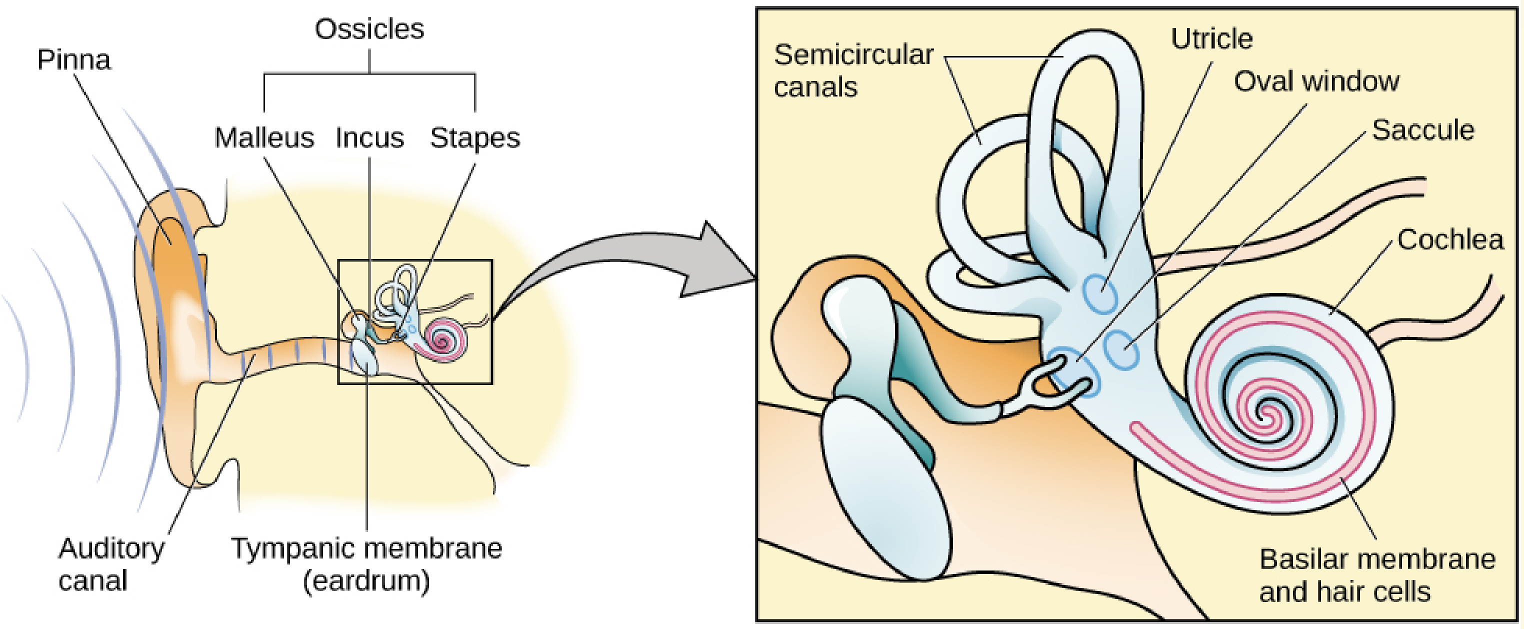
We perceive depth in a 2-D image through monocular cues like converging parallel lines. *(credit: Marc Dalmulder)*

Hearing



The ear: outer (pinna, tympanic membrane), middle (ossicles: malleus, incus, stapes), inner (cochlea, basilar membrane).

How We Hear



Sound waves → tympanic membrane → ossicles → oval window → cochlear fluid → **hair cells** → auditory nerve → brain (auditory cortex in temporal lobe)

Hair cells are the auditory receptor cells. Their mechanical activation generates neural impulses.

Pitch Perception

Two theories:

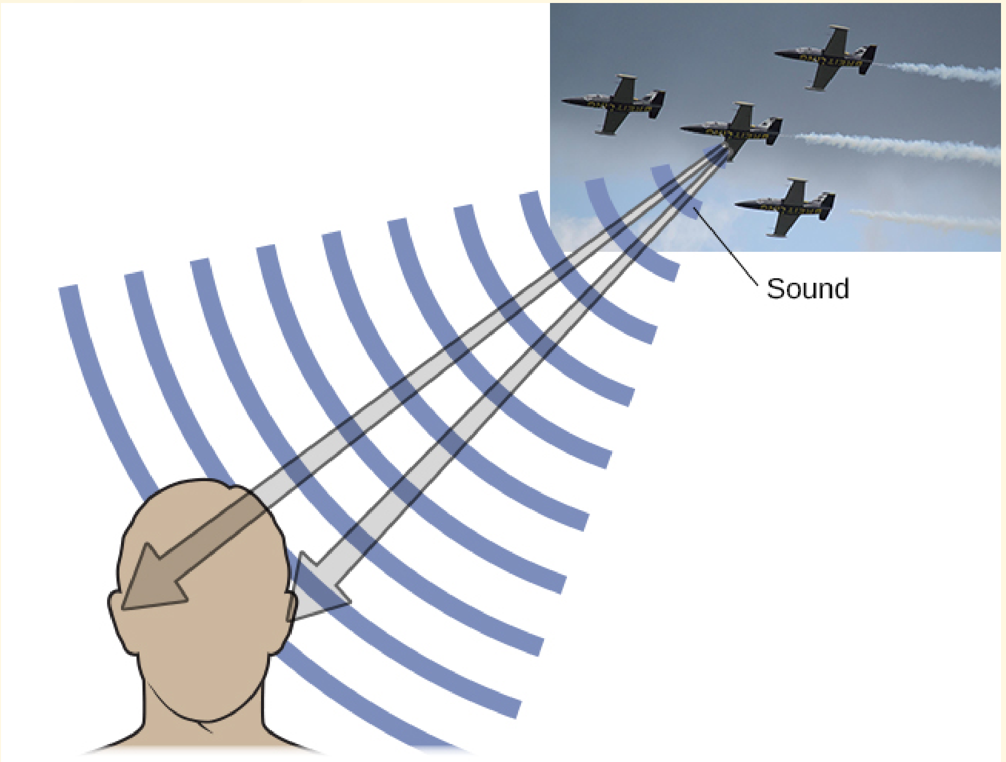
Temporal theory — frequency coded by firing rate of hair cells. Limited to ~4,000 Hz.

Place theory — different parts of basilar membrane respond to different frequencies. Base = high frequency; tip = low frequency.

Both apply: up to ~4,000 Hz, both rate and place contribute. Higher frequencies rely on place alone.

Sound Localization

Sound localization uses both monaural and binaural cues.



- **Monaural cues**: pinna shape → above/below and front/behind
- **Interaural level difference**: louder in nearer ear
- **Interaural timing difference**: tiny delay between ears → left/right location

Hearing Loss

Conductive hearing loss — failure in vibration of eardrum/ossicles; treatable with hearing aids.

Sensorineural hearing loss — failure in neural transmission from cochlea to brain; most common form.

Causes: aging, noise exposure, trauma, infections, medications.

Cochlear implants — bypass hair cells and directly stimulate the auditory nerve.

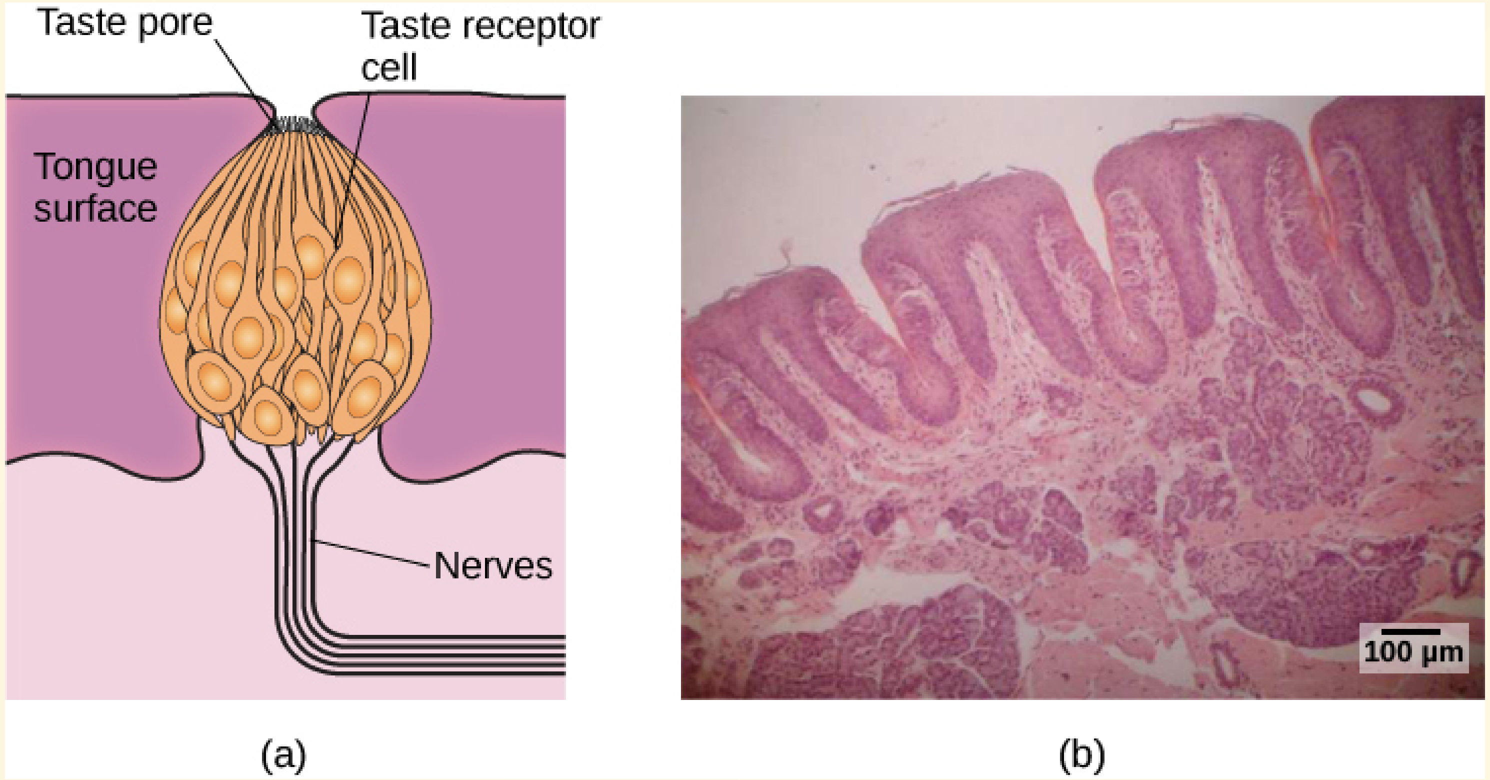
The Other Senses

Chemical Senses: Taste (Gustation)

At least **6 taste qualities**: sweet, salty, sour, bitter, **umami**, and fatty.

- Taste buds on the tongue detect molecules dissolved in saliva.
- Taste information → medulla → thalamus → limbic system → **gustatory cortex** (frontal-temporal junction)
- Taste buds regenerate every 10–14 days.

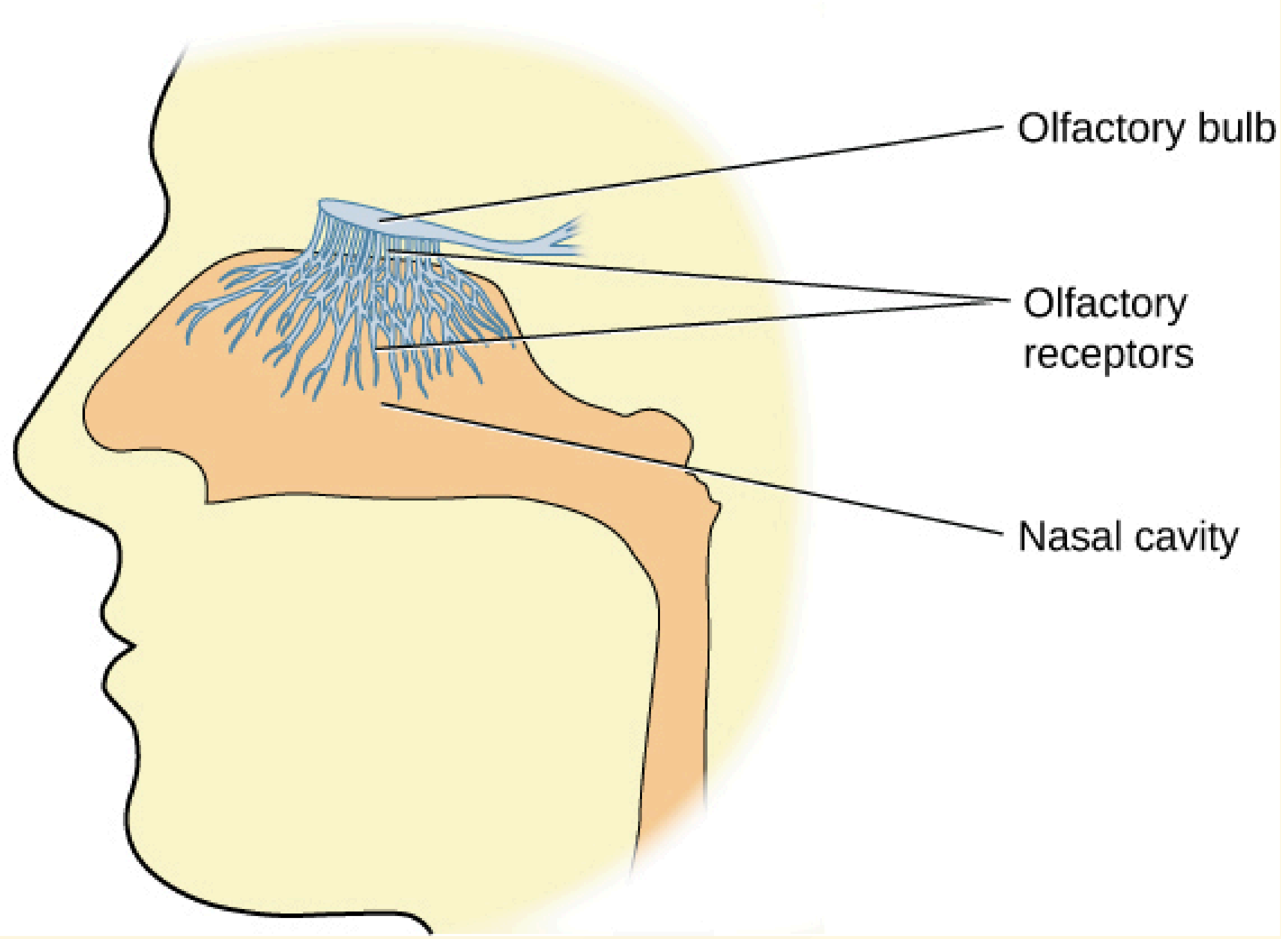
Taste Bud Anatomy



(a) Taste receptor cells in a taste bud.
(b) Microscopic view of tongue surface.

Chemical Senses: Smell (Olfaction)

- Olfactory receptors in the nasal mucous membrane detect airborne molecules.
- Signals go to the **olfactory bulb** → limbic system and **primary olfactory cortex**.
- Dogs have 800–1,200 functional olfactory receptor genes vs. ~400 in humans.
- **Pheromones** — chemical signals between individuals (prominent in many species).
- Taste and smell work **together** to produce flavor perception.



Olfactory Receptors

Hair-like olfactory receptor extensions protrude into the nasal mucous membrane to bind odor molecules.

Touch: Somatosensation

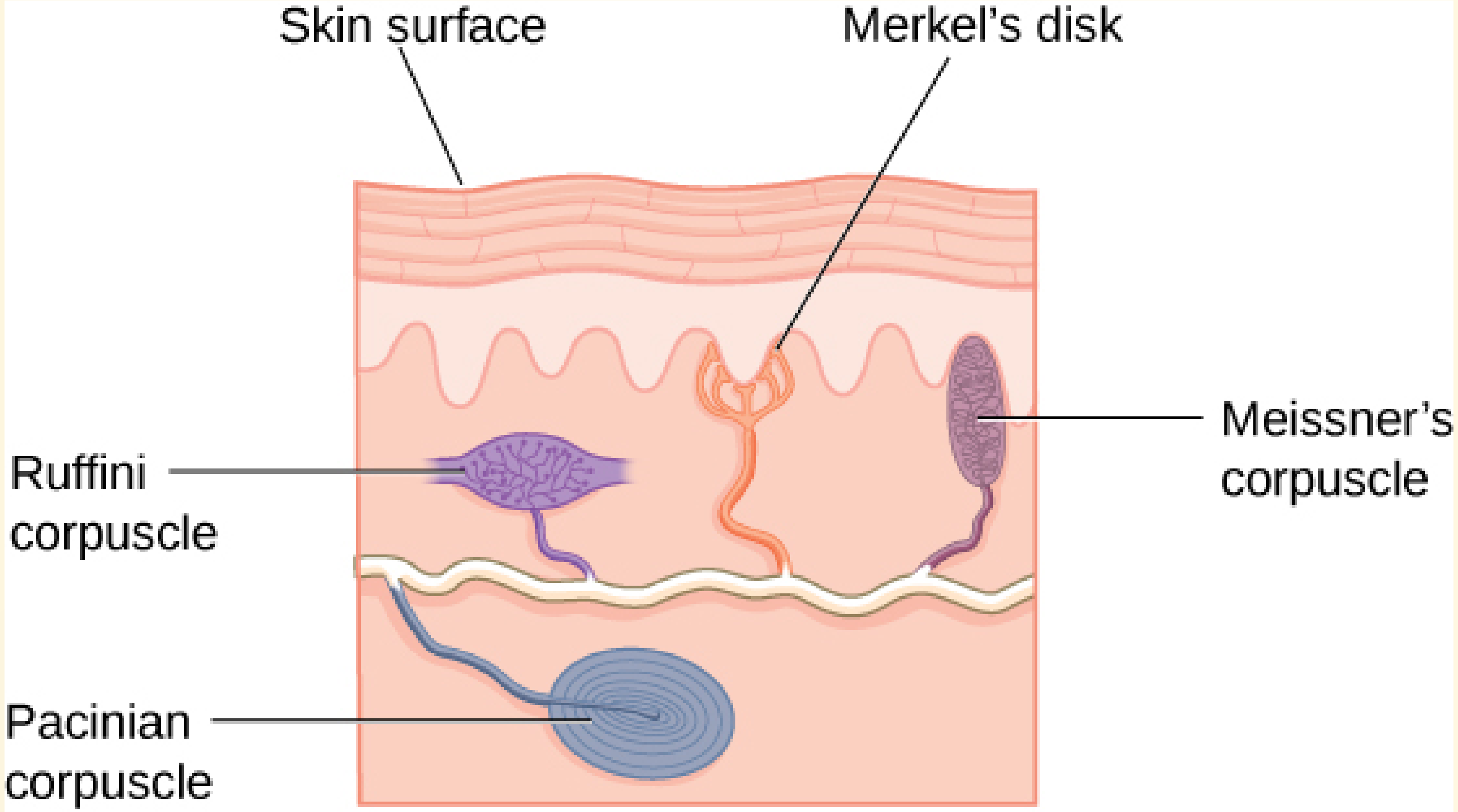
Four main types of skin receptors:

Receptor	Responds to
Meissner's corpuscles	Pressure, low-frequency vibration
Pacinian corpuscles	Transient pressure, high-frequency vibration
Merkel's disks	Light pressure
Ruffini corpuscles	Stretch

Free nerve endings also detect **temperature (thermoception)** and **pain (nociception)**.

Skin Receptors

Multiple receptor types in the skin respond to different touch stimuli. Signals travel via spinal cord to **somatosensory cortex** (postcentral gyrus).



Pain (Nociception)

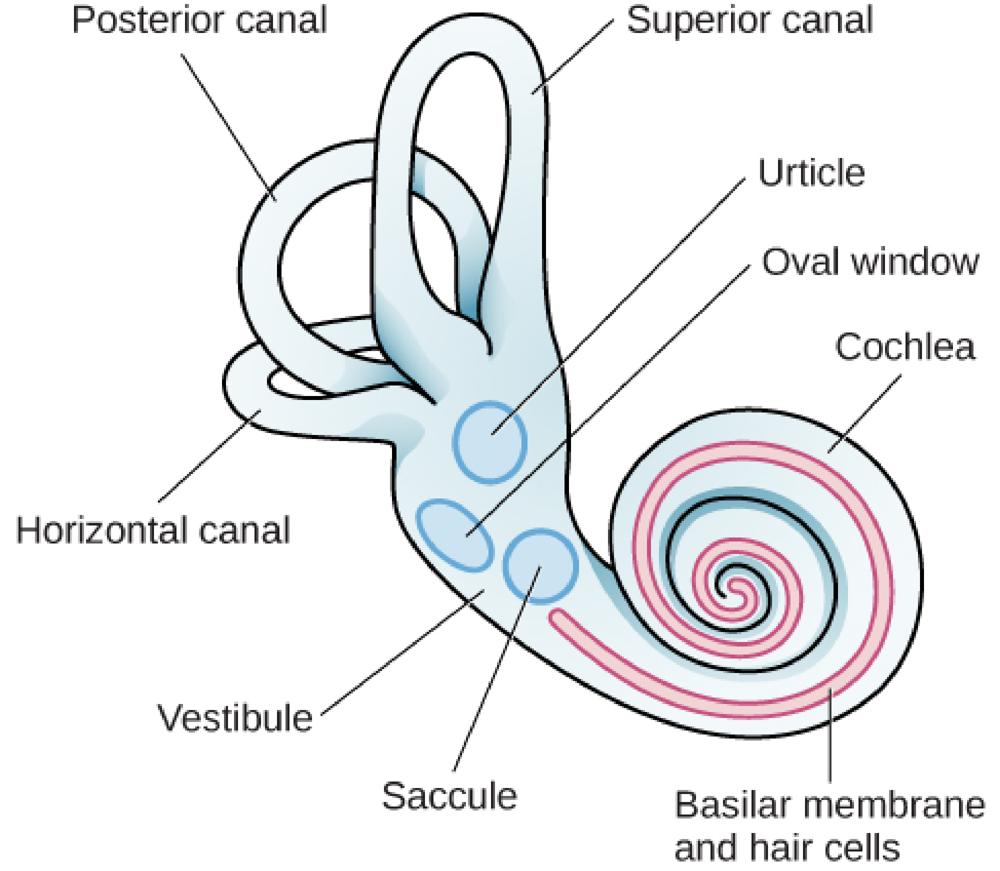
Pain is adaptive — signals injury and motivates avoidance.

- **Inflammatory pain** — signals tissue damage
- **Neuropathic pain** — amplified signals from damaged neurons

Nociceptors: free nerve endings located in the skin and other tissues that respond to harmful stimuli: mechanical, thermal, chemical. **Congenital insensitivity to pain (congenital analgesia)** — extremely rare; individuals sustain severe unnoticed injuries and have shorter life expectancies.

Vestibular, Proprioceptive & Kinesthetic

Vestibular organs (utricle, saccule, semicircular canals) next to the cochlea in the inner ear.



- **Vestibular sense** — balance and head/body orientation
- **Proprioception** — perception of body position
- **Kinesthesia** — perception of body movement through space

Gestalt Principles of Perception

Gestalt Psychology

Founded by **Max Wertheimer** (early 20th c.), with Köhler and Koffka.

"The whole is different from the sum of its parts."

The brain organizes sensory input into meaningful perceptions in **predictable ways** — these are the Gestalt principles.

Key idea: perception is an active construction, not a passive recording.

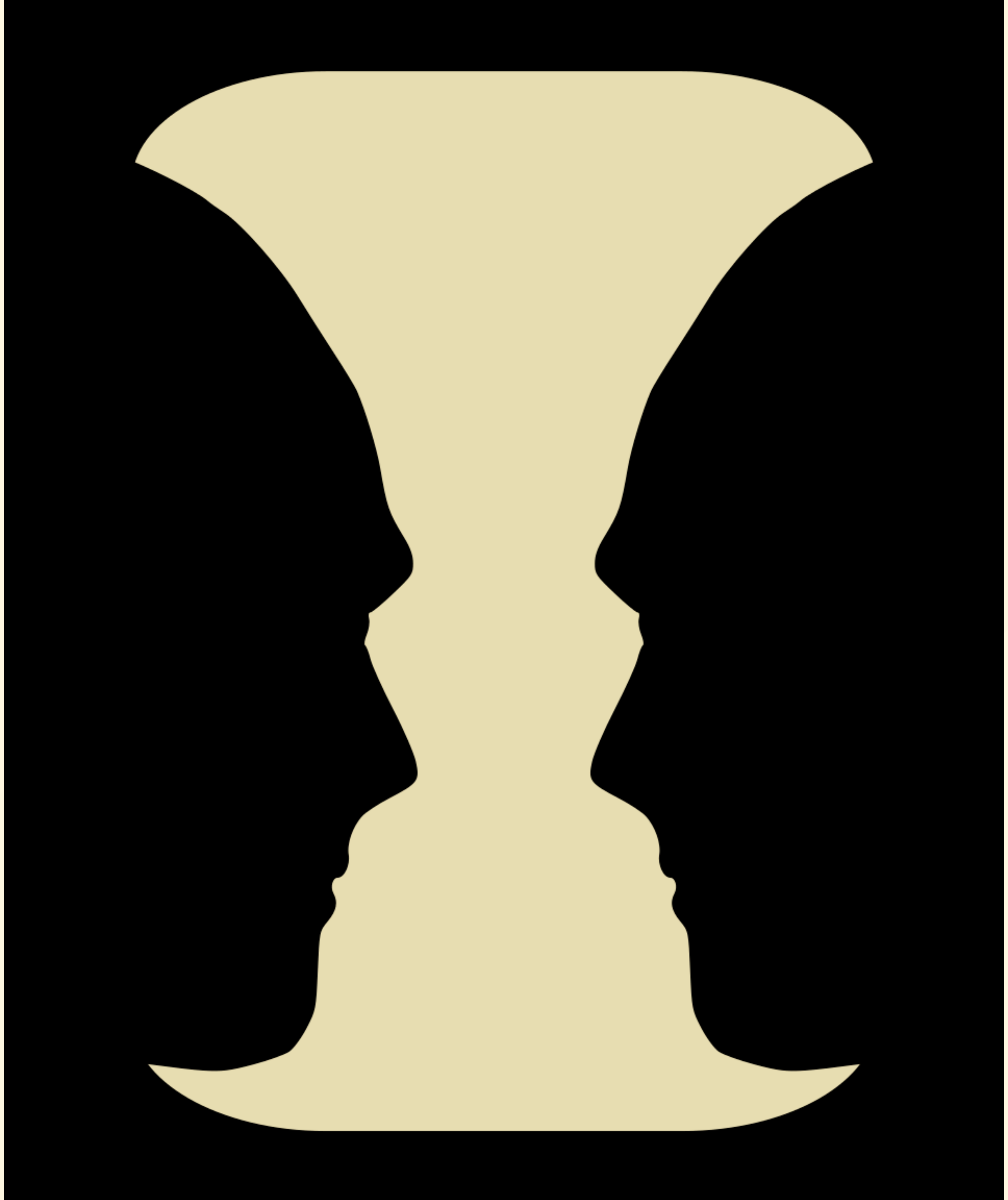
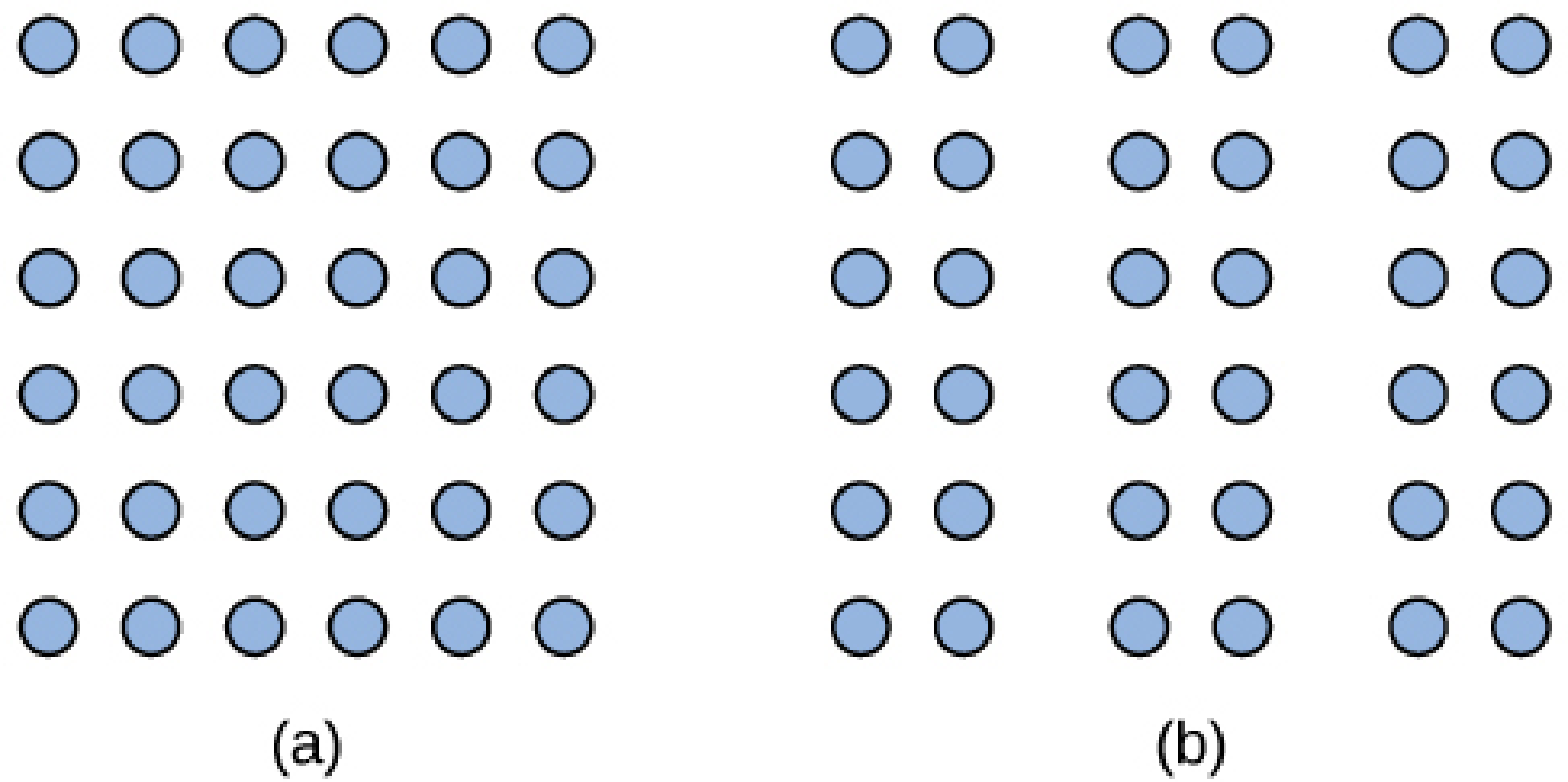


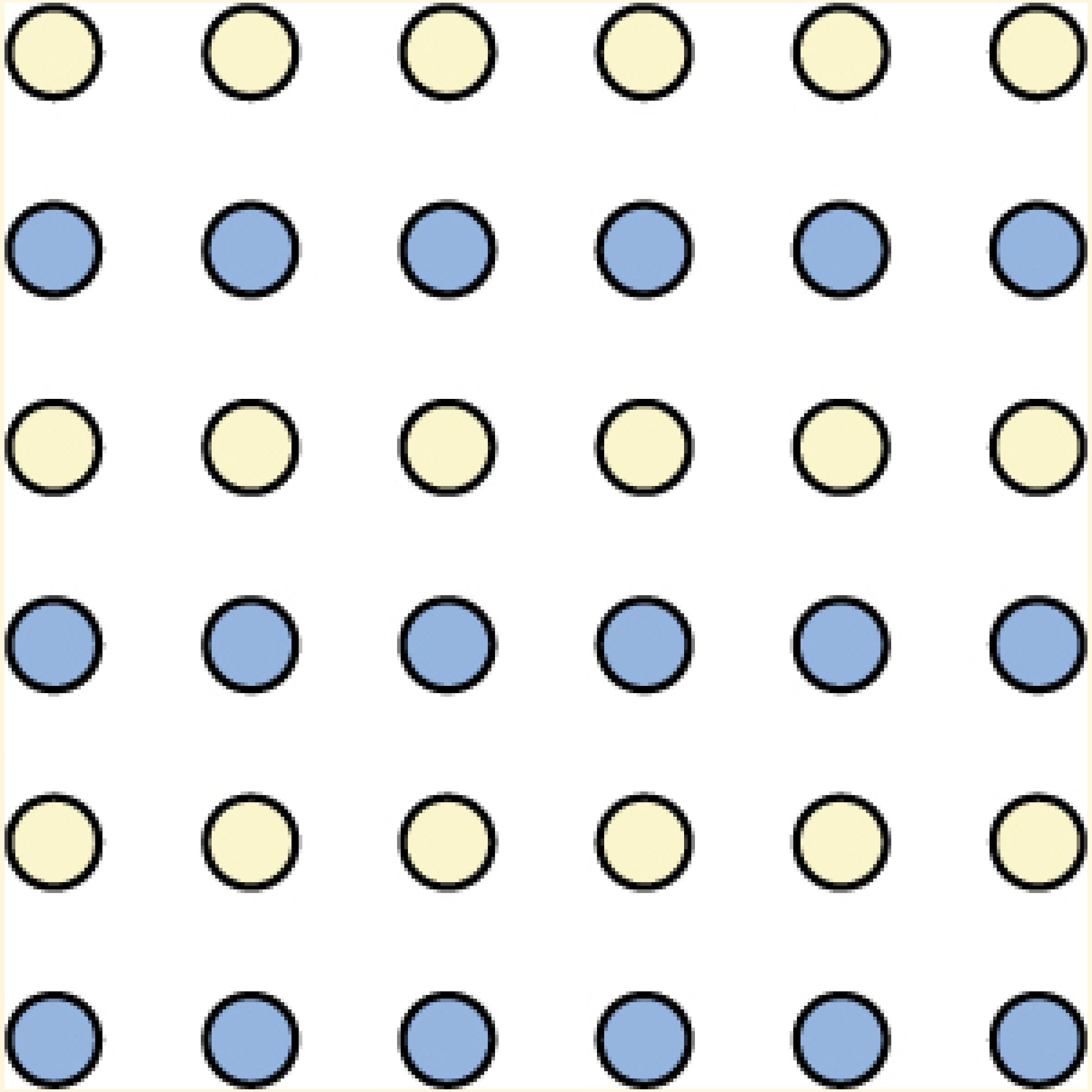
Figure-Ground Relationship

The Rubin vase: perception alternates between a vase (white figure) and two faces (black figure), depending on what is perceived as figure vs. ground.

Proximity



Proximity: things close to each other tend to be grouped together. Left: one block; right: three columns.



Similarity

Similarity: like elements are grouped together. We perceive alternating colored rows rather than a grid.

Good Continuation & Closure

Good continuation: we perceive two overlapping lines, not four separate line segments.
Closure: we perceive complete shapes (circle, rectangle) even when parts are missing.

Perceptual Set & Bias

Perceptual hypotheses — educated guesses when interpreting sensory information, shaped by:

- Personality and prior experience
- Expectations and verbal priming
- Cultural background and implicit biases

Research shows implicit racial bias can affect perception of objects (e.g., misidentifying non-weapons as weapons when paired with racialized images).

Principles of Multisensory Integration

- **Temporal principle**: stimuli from different modalities are more likely integrated if they occur at the same time.
- **Spatial principle**: stimuli are more likely integrated if they occur at the same location.
- **Inverse effectiveness principle**: integration is stronger when individual unisensory signals are weak.

Multisensory Illusions

- **McGurk effect**: mismatched audio and visual speech cues produce a third, different perception.
- **Ventriloquist effect**: sound source perceived at the visual location (puppet's mouth) rather than the actual acoustic source.
- **Rubber hand illusion**: synchronous touch of a fake and real hand induces sense of ownership of the fake hand.

Summary

- Sensation ≠ Perception; attention, adaptation, signal detection
- Waves: amplitude → intensity; frequency → pitch/color
- Vision: rods/cones, color theories, depth cues
- Hearing: hair cells, pitch theories, sound localization
- Other senses: taste, smell, touch, pain, balance
- Gestalt: figure-ground, proximity, similarity, closure

Quiz and Sources

- Quiz (multiple choice) is already set in UNIPA
- Time: 2026-05-26 ~ 2026-06-8
- Only two chances are permitted
- 25 questions, 50 minutes

Main source:
OpenStax Psychology 2e, Chapter 5 (Spielman, Jenkins & Lovett, 2020).

Next lecture

- Focus on multimodal processing, i.e., computation model
- We will practice coding as the task, e.g., [Multibench]() and [Nkululeko]()
- Multibench can be used for multimodal processing, text, image, audio; while Nkululeko currently only support fusion of text and audio (from speech).

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[FEATS]
# Combine BERT linguistic features with OpenSMILE acoustic features
type = ['bert', 'os']
os.set = eGeMAPSv02
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